Unit : 1

Unit 1. Web Server Concept
1.1. Introduction to Web Server
1.2. Architecture of web server
1.3. Concept of Dynamic Content
1.4. Using control flow to control dynamic content generation
1.5. Concept of Architecting Web Application

Web server

Web server is a computer where the web content is stored. Basically web server is used to host the web sites but there exists other web servers also such as gaming, storage, FTP, email etc

Website is a collection of web pages while web server is a software that respond to the request for web resources.

Web server respond to the client request in either of the following two ways:

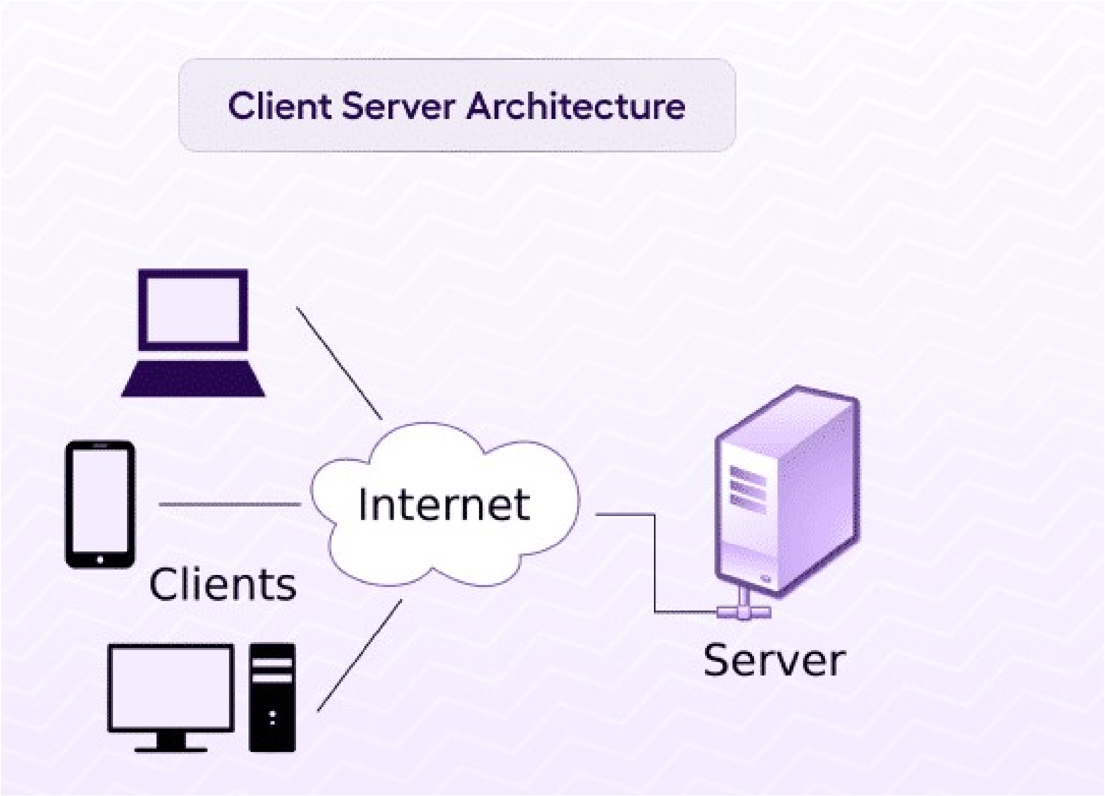
Sending the file to the client associated with the requested URL

Generating response by invoking a script and communicating with database.

Web Browser:
Web browser is an application software that allows us to view and explore Information on the web. User can request for any web page by just entering URL in to address bar.

Web Browser can show text, audio, video, animation and more. It is the Responsibility of web browser to interpret text and commands contained in the web page.

Client server architecture



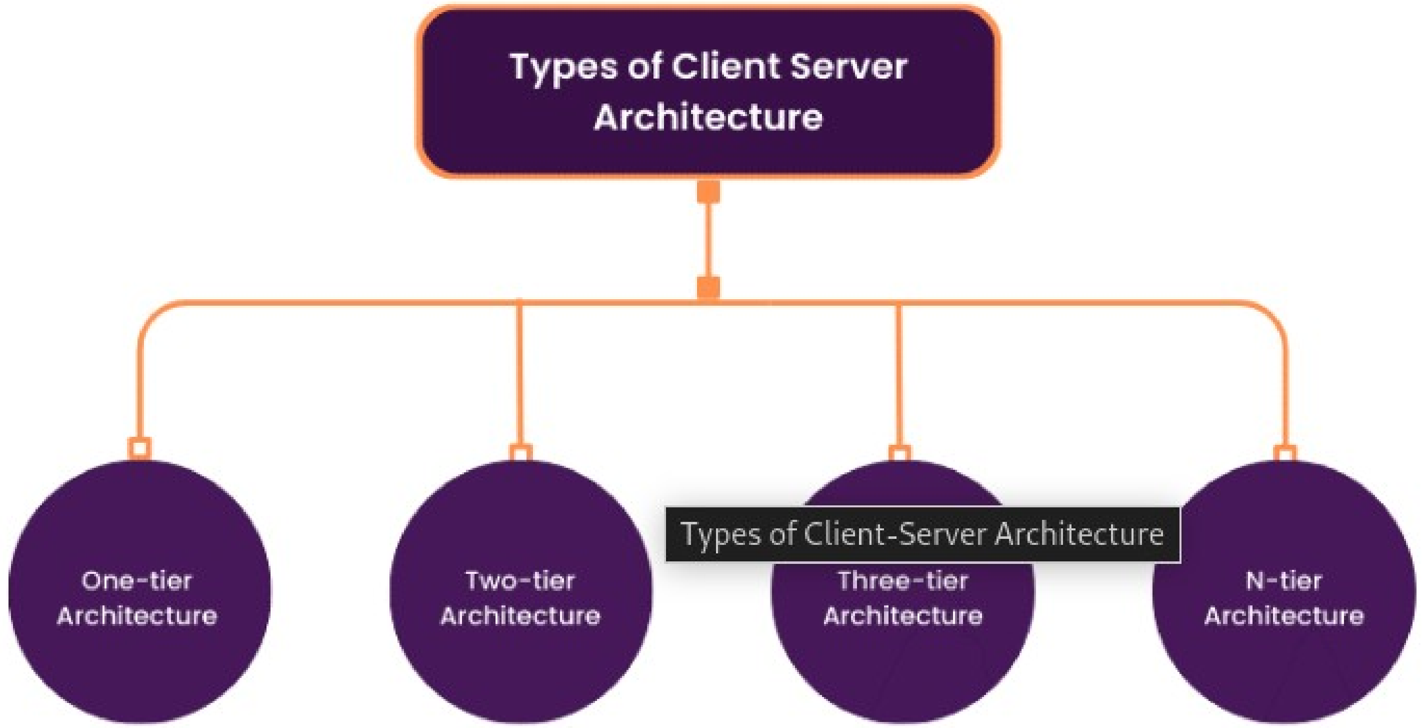
The main players in this architecture are clients and servers. They enable the communication process. To understand how client-server architecture works, it is important to understand the communication process between the two factors.

Requests: Clients send requests to the server to notify it of events, like a user trying to log in with his credentials or requesting data, such as files.

Response: The server responds to the client's requests in the form of messages. This might be the result of an authentication.

Service: A service is a specific job that the server makes accessible to the client for execution, such as downloading an image.

Types of client server architecture



There are different types of Client-Server Architecture, depending on how many tiers or layers are involved in the communication process. Some of the common types are:

- 1) One-tier Architecture

  A self-contained application on a single platform. In one-tier architecture, the client, server, and database are all on the same machine. The client handles user interaction and business logic, the server provides services like data storage and processing, and the database manages data. While simple and popular for small apps, this architecture is rarely used in production because it doesn't meet most system requirements.

  All tier In one machine

  2) Two-tier Architecture

- This basic Client-Server Architecture involves direct communication between the client and server without an intermediate layer. The client manages the User Interface (UI) and business logic, while the server handles data storage and processing. An example is a web browser requesting pages from a web server, which responds with HTML files. It's easy to implement but has drawbacks like low scalability, high network traffic, and security risks.

  Two different tire in two different layer

  -client

  -server

Three-tier Architecture

- A more complex client-server setup with an intermediate layer (usually middleware or an application server) that handles business logic, acting as a bridge between the client and server. The client deals with the user interface (UI), while the server manages data storage. An example is an online banking system where the client is a web browser, the middleware checks transactions, and the server stores account data. This architecture improves scalability, performance, and security but increases complexity and cost.

Three different tier

-client

-server

-application

- N-tier Architecture
- N-tier architecture consists of more than one architecture: This includes examples like
- Used for large systems with microservices, APIs, load balancers, etc.

Architecture of Web Server

What is Web Server Architecture?

A Web server architecture establishes the arrangement, elements, and setups that are essential for its effective operation and the provision of web-oriented services to users. This constitutes a pivotal facet of web server administration, exerting substantial influence over the server’s performance, dependability, and safeguarding.

The two main approaches to web server architecture are the concurrent approach and the single-process-event-driven approach.

Concurrent Approach:
In the concurrent approach, the web server creates a separate process or thread for each client request.
Each process or thread handles one request at a time, allowing multiple requests to be processed simultaneously.

For example, suppose a web server receives five client requests simultaneously. In the concurrent approach,
the server will create five separate processes or threads to handle each request concurrently.
This approach is often used in traditional web servers like Apache, which can handle a large number of concurrent connections.

Single-Process-Event-Driven Approach:
In the single-process-event-driven approach, the web server uses a single process or thread to handle all client requests. The server waits for events (e.g., new connection requests, and incoming data) and handles them one at a time in a non-blocking manner.

For example, suppose a web server receives five client requests simultaneously. In the single-process-event-driven approach, the server will use a single process or thread to handle all five requests one at a time, in a non-blocking manner. This approach is often used in modern web servers like Node.js, which are designed to handle large numbers of lightweight connections efficiently.

Dynamic content:

Dynamic content refers to web elements that change in real time based on a user's behavior, preferences, location, or past interactions.content. Unlike static content, which remains the same for all visitors, dynamic content personalizes the user experience by displaying relevant product recommendations, location-based offers, or customized messaging..

- Some common examples of dynamic content include:
- Personalized emails – Content dispays to individual users based on their interactions.
- Targeted ads – Ads that adjust based on browsing history and preferences.
- Customized product offers – E-commerce sites showing relevant products based on previous searches.
- Geo-data updates – Websites displaying location-based news

- Key Characteristics:

  Personalization:
- Dynamic content can tailor experiences based on user data, location, or preferences.

  Real-time Updates:

  Content can be updated instantly without requiring a full page refresh.
- Interactivity:

  Dynamic content allows for interactive elements like forms, pop-ups, and sliders.
- Engagement:

  By offering personalized and relevant content, dynamic websites can increase user engagement.
- Examples:
- Product Recommendations: Suggesting related products based on browsing history.
- Personalized Messages: Displaying a greeting with the user's name or offering personalized discounts.
- Location-Based Content: Showcasing local events or services based on the user's location.
- Live Updates: Displaying real-time data like stock prices or weather updates.

control flow to control dynamic content generation

Control flow means using programming statements like if, else, for, or while to decide how content should be generated or displayed—based on user input, data, or conditions.

- When building dynamic content, control flow helps us customize the output on a webpage.
- It allows the program to:

-Show different content to different users

-Loop through data (like a list of products or messages)

-Display content only if a condition is true

Concept of Architecting web application:

A web application is a software or program that runs on a web browser and is accessed via the internet (or sometimes an intranet). Unlike traditional desktop applications, users don’t need to install anything, just open a browser like Chrome or Firefox and use it.

- Runs on a web server
- Accessed through URLs
- Uses HTML, CSS, JavaScript for frontend
- Uses backend languages like Python, PHP, or Node.js for logic
- Can connect to a database to store and manage data

Presentation Layer:

- This layer handles the user interface and how users see information. It includes design, user experience, and interactions. Developers use front-end frameworks like React, Angular, or Vue.js to build responsive and dynamic interfaces. The presentation layer displays content and processes user inputs, ensuring a smooth and visually appealing experience.

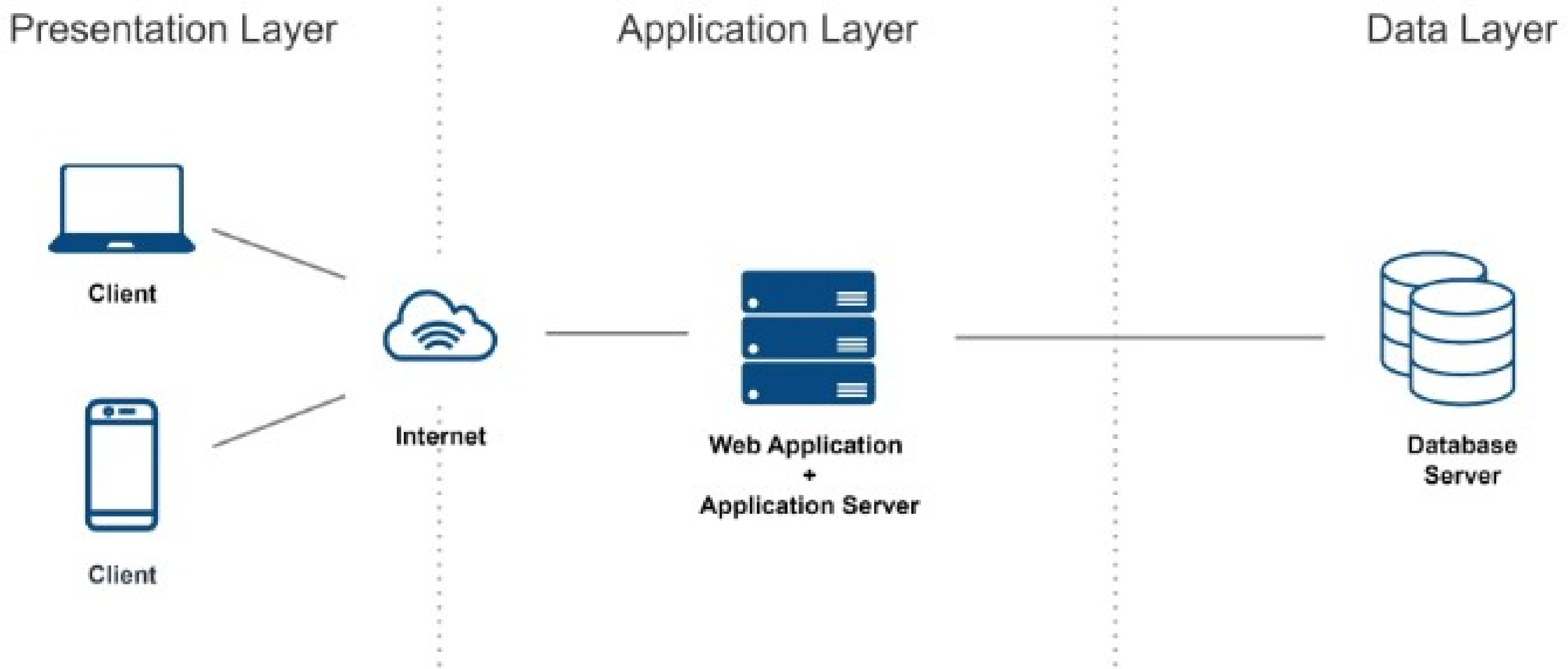
- Application Layer:

  The application layer, also called the business logic layer, handles the main functions of a web application. It processes user requests, applies business rules, and manages how the application works. This layer connects the presentation layer with the data access layer, ensuring the application runs as expected.

- Data Access Layer:

  The data access layer connects the application to the database or storage system. It handles data retrieval and storage while ensuring secure and efficient access. This layer hides the details of data storage from the rest of the application, making it easier to update or change the storage system without affecting the application's main logic.

Web Application Architecture



Static Server and Dynamic Web server

Feature	Static Web Server	Dynamic Web Server
Content Type	Delivers fixed, pre-written files	Generates content on-the-fly
Content Change	Same content for all users	Changes based on user input/data
Programming	No server-side scripts	Uses server-side languages (PHP, Python, etc.)
Database Support	Not needed	Usually connected to a database
Speed	Very fast	Slower due to processing
Example Technologies	HTML, CSS, JS, Apache, Nginx	PHP, Flask, Node.js, Django + Apache/Nginx
Use Case	Simple websites, portfolios	Web apps, login systems, e-commerce

Feature	Static Webpages	Dynamic Webpages
Content	Fixed, same for all visitors	Personalized or changing content
Programming	HTML, CSS, basic JS only	HTML + PHP, Python, Node.js (server-side logic)
Database	Not used	Frequently uses databases
User Interaction	Limited or none	Interactive and responsive
Example	"About Us" page, static blog	Login page, dashboard, shopping cart
Speed	Loads instantly	Slight delay due to processing

Old Questions:

- Define web server? Explain the architecture of web server?
- Explain the concept of dynamic content